DO NOW — RECALL

Lesson 149 · Client–Server and Peer-to-Peer Network Mo

Name: _____ Class: _____ Date: _____

1. Which device connects several wireless users to an existing school LAN?

2. Which storage type is volatile?

3. Which layer in TCP/IP handles IP routing?

4. What data structure follows FIFO?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A wireless access point.

2. Model answer

RAM.

3. Model answer

The internet layer.

4. Model answer

A queue.

By the end, you can…

• Explain how client–server and peer-to-peer models divide responsibility and resources.

• Compare reliability, administration, cost, security and scalability in context.

• Recognise hybrid systems that use central identity but peer-to-peer data transfer.

Vocabulary — know these terms

client–server — Managed service provided by server to requesting clients.

peer-to-peer — Peers directly share services/resources.

centralisation — Services controlled from a smaller set of authorities.

scalability — How well a system handles growth in load/users.

NEW LEARNING

Read, annotate and explain

Client–server

Many clients request services from a managed central server.

Peer-to-peer

Each participating peer may offer and consume resources.

Select architecture

Balance availability, management, trust and workload distribution.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A school stores assessment marks requiring access control, backups and a shared authoritative copy. Recommend a model.

Teachers send requests

Identify the facts and make an initial prediction.

Central marks database

Explain the mechanism or trace the intermediate stage.

Access control + backup

Client–server suits managed central data, controlled permissions, auditing and backups. A server failure can affect many clients, so redundancy and resilient backups matter. Peer-to-peer file sharing is harder to manage consistently.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define client, server and peer in the context of a network service. [6]

Question 2

Explain two benefits and one limitation of client–server for a school's student database. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare P2P and client–server for distribution of a large video archive, including integrity checks. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A network using client–server can never contain any peer-to-peer activity.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.